REAL-TIME CYBER THREAT INTELLIGENCE

CyberShield

One screen. Every signal that matters — collected, ranked and explained as it happens.

CyberShield is a self-contained cyber threat intelligence dashboard. It continuously harvests nineteen research streams from across the security internet, cross-checks every story against the official known-exploited vulnerability catalog, and distills the noise into a ranked, explainable operational picture — complete with an AI analyst that turns any headline into a structured threat brief. This manual explains what CyberShield is, what it is for, and exactly how it works.

19 LIVE RESEARCH STREAMS	**3D** GLOBAL THREAT MAP (WEBGL)
AI ANALYST VERDICTS	**KEV** EXPLOIT TRACKING

// SECTION 01

What is CyberShield?

CyberShield is a **real-time cyber threat intelligence dashboard**: a single, always-on screen that turns the constant churn of security reporting into an ordered, prioritized picture. Instead of juggling a dozen feeds, bookmarks and mailing lists, an operator opens one page and sees what is happening across the global threat landscape right now — already collected, deduplicated, ranked by severity, and enriched with context.

Everything runs on its own server layer. The dashboard is fully self-sufficient: it needs no external database service, no paid API subscriptions and no third-party aggregation account. Feeds are fetched from the public internet at request time, cached intelligently in memory, and served through a small set of purpose-built API routes. Drop it on any host and it simply works — the first visit triggers a full harvest of all sources, and every visit after that is instant.

It is also designed to be **honest about its own state**. Pipeline status tells you when data is fresh and when a background refresh is running; the known-exploited catalog carries a visible sync indicator; and the AI analyst clearly labels any verdict produced by its deterministic fallback engine rather than a live model. Nothing on the screen pretends to be more current or more certain than it actually is.

WHAT IT IS	A live, self-updating threat intelligence dashboard for teams, researchers and enthusiasts
WHAT IT SHOWS	19 aggregated research streams, a severity-ranked threat board, KEV exposure tracking and a 3D global attack map
WHAT IT EXPLAINS	Per-story analyst verdicts: threat class, severity rating, impact summary and recommended immediate actions
WHAT IT NEEDS	Any Node or bun host. No external database service, no paid APIs, no credentials required to run
HOW IT IS BUILT	Next.js 16 App Router + TypeScript server layer, WebGL threat map, memory-first caching with optional SQLite archive

// SECTION 02

What is it for?

CyberShield is built for anyone who needs **situational awareness without an enterprise threat-intelligence budget**. Security teams use it as a shared wall display; researchers use it to watch campaigns develop; educators use it to demonstrate how a real intelligence pipeline behaves. In every case the value is the same: the signal-to-noise work is done once, server-side, so that looking at the screen is enough.

01

SOC & Operations Walls

Pin it to the big screen in the war room. The ranked board keeps attention on the stories that actually matter, and known-exploited hits surface immediately instead of hiding in a list.

02

Vulnerability Watch

The official Known Exploited Vulnerabilities catalog is tracked next to live reporting, with an exposure gauge — so "is this being exploited in the wild?" is answered at a glance.

03

Research & Due Diligence

Investigate any story in one click. The analyst verdict expands a headline into threat class, severity, impact and concrete immediate actions — a starting brief, not just a link.

04

Training & Education

A safe, realistic environment for teaching how intel pipelines work: sources, parsing, deduplication, ranking, enrichment and graceful fallbacks — all visible in one product.

05

Incident Context

When something breaks, check within seconds whether the wider internet is already reporting related campaign activity — before you write the first incident note.

06

Portfolio & Demonstration

A fully working, deployable product that demonstrates modern full-stack engineering: streaming data, WebGL visualization, AI integration and zero-infrastructure operations.

// SECTION 03

How it works — the intelligence pipeline

Collection happens **server-side, on every refresh**. Nineteen curated streams — vendor advisories, CERT bulletins, research labs and security aggregators — are polled by the server, parsed from their native XML first. If a source refuses a direct fetch, the request falls through a public RSS-to-JSON converter and then a server-side relay, so a single blocked host never blanks the picture.

Each story is then **normalized and enriched**: the best available thumbnail is extracted through a hotlink-safe image relay, per-story analyst briefs are distilled, duplicates across sources are collapsed, and everything is ranked by severity, recency and known-exploitation. The result is cached in memory and served as one aggregated snapshot.

01 **COLLECT**
19 research streams polled server-side — direct XML first, public converter and server relay as fallback

02 **PROCESS**
Parse · extract thumbnails · distill analyst briefs · dedupe across sources · rank by severity, recency and KEV presence

03 **CACHE**
Memory-first snapshots — fresh for minutes, instantly available when stale, refreshed in the background

04 **SERVE**
One aggregated snapshot feeds the dashboard, the ranked board, the 3D map and every connected client

LAYER	ENDPOINT / STORE	FRESH WINDOW	BEHAVIOR WHEN STALE
Intel snapshot	/api/news · memory	4 minutes	Serves the stale snapshot instantly while a background refresh runs — for up to 30 minutes
KEV catalog	/api/cisa · memory	Session-long	Serves the cached catalog and re-syncs from the official source or its 1:1 mirror
Thumbnails	/api/img · LRU (250)	30 minutes	Re-fetches through the hotlink relay on demand; private-network targets are blocked
Verdict archive	/api/brief · SQLite	Permanent	Instant re-open of past investigations; REANALYZE forces a fresh verdict
Converter relay	/api/rss2json · memory	4 minutes	A short TTL shields the public converter from request storms

TABLE 01 — CACHING LAYERS AND STALE-HANDLING BEHAVIOR

The fallback philosophy runs through the whole pipeline: **every external dependency has a second path.** Hotlink-protected thumbnails are proxied with a same-origin referer and strict image validation; the KEV catalog has a mirror; feed conversion has a relay.

The dashboard keeps rendering even when individual sources misbehave — a degraded source simply drops out of the ranking until it recovers.

// SECTION 04

How it works — analysis & context

Two context systems sit on top of the raw stream. The first is **ground truth**: the official CISA Known Exploited Vulnerabilities catalog. The full catalog is fetched server-side and cached, with an auto-synced mirror as fallback, so the multi-megabyte download never blocks the hot path. Stories referencing cataloged CVEs are ranked up, and the exposure gauge at the top of the board summarizes the current exploited-vulnerability load at a glance.

The second is the **AI Analyst**. Clicking any story opens the investigation drawer and produces a structured, SOC-grade threat brief: threat class, severity rating, what happened, why it matters, and immediate recommended actions. The analysis walks a three-layer chain so it can never dead-end:

1	**LIVE MODEL** Configured credentials — any OpenAI-compatible provider — generate the full verdict in real time.
2	**FORWARDED AUTHENTICATION** If the hosting gateway injects fresh per-user credentials into the request, those are used instead of a possibly stale file token.
3	**OFFLINE ANALYST** A deterministic heuristic brief distilled from the server-side intel pipeline — lede, highlights, CVEs, organizations, severity. Clearly marked as degraded, and never archived in place of a live verdict.

Completed verdicts are **archived in SQLite**, keyed by the normalized article URL, so re-opening a past investigation is instant. Degraded fallback verdicts are not archived — a later live re-analysis replaces them. The feature is therefore fully usable out of the box, and becomes sharper the moment real model credentials are provided.

// SECTION 05

The experience

CyberShield is deliberately theatrical about security operations — **the interface behaves like a mission system, not a blog list**. Every dramatic element is backed by a real function: the animations cover real sync time, the voice marks a ready state, and the map renders the live feed itself.

01

Threat-Radar Boot Sequence

While the first 19-source sync runs, a radar calibration animation takes over the screen. Cold starts feel intentional — the system visibly comes online while real work happens underneath.

02

JARVIS Voice Welcome

A synthesized welcome voice greets visitors when the dashboard reaches readiness. It plays automatically where browser autoplay policy allows, and on first interaction everywhere else.

03

3D Global Threat Map

Continents rendered as a dot-matrix from real coastline data. Live stories trace attack arcs across the globe with impact rings — the map is driven by the same feed that powers the board.

04

Live Feed Columns

Dense, scannable columns with real thumbnails, source tags, keyword chips and age stamps. Thumbnails are fetched through the hotlink relay, so images that break elsewhere render here.

05

Ranked Threat Board

Severity-ranked stories with category filters and keyword tracking, anchored by the KEV exposure gauge. The ranking is computed server-side from the same intelligence snapshot.

06

Investigation Drawer

One click turns any headline into a structured analyst brief. A world clock, live ticker, sound toggle and a fully responsive mobile layout keep the whole room — and every device — oriented.

// SECTION 06

Under the hood

The stack is intentionally boring where it should be boring and modern where it pays off: a typed, server-rendered application with a small API surface, hardware-accelerated visualization, and memory-first persistence. Nothing needs to be provisioned to run it.

LAYER	TECHNOLOGY	ROLE
Application	Next.js 16 · App Router	Server layer + dashboard UI, standalone build output
Language	TypeScript (strict)	End-to-end typing across routes, libraries and components
Styling	Tailwind CSS	Cyber palette, responsive layout system, dark-first design

LAYER	TECHNOLOGY	ROLE
3D Map	Three.js · WebGL	Dot-matrix Earth, attack arcs, impact rings, real land data
Motion	GSAP	Boot choreography, drawer and panel transitions
Persistence	Prisma + SQLite	Optional verdict archive; every other cache is memory-first
Runtime	bun	Development, dependency management and standalone serving

TABLE 02 — TECHNOLOGY STACK

ENDPOINT	PURPOSE	RESILIENCE
/api/news	Aggregated 19-stream intelligence snapshot	Three-layer fetch fallback, dual-TTL cache
/api/brief	AI analyst verdicts with archival	Three-layer model chain, SQLite archive
/api/cisa	Known Exploited Vulnerabilities relay	Official source + mirror, memory cached
/api/img	Hotlink-safe thumbnail relay	Referer handling, validation, LRU cache
/api/rss2json	RSS-to-JSON conversion relay	TTL cache shields the public converter

TABLE 03 — API SURFACE

Long-running routes declare **60-second execution ceilings** in the route files themselves, so the initial full sync is never cut off by conservative platform defaults on metered hosting tiers. The build produces a standalone server bundle, which is what makes container deployments so small and predictable.

// SECTION 07

Deployment & operations

CyberShield ships as **a single deployable unit**. Because caching is memory-first and the only database touchpoint is the optional verdict archive, there is no infrastructure to provision: no managed database, no message queue, no worker fleet. Three deployment paths cover every realistic budget, and all of them support custom domains.

PATH A

Vercel — Free Tier

Push the project to GitHub, import it on Vercel, deploy. Framework preset auto-detects; no environment variables are required. Function time limits are already baked into the routes, so the first sync survives the free plan's defaults.

PATH B

Docker — Any Host

A self-contained container image: build it, run it on port 3000, done. The included SQLite file makes the verdict archive work inside the container, giving the full feature set on any VPS or container platform.

PATH C

Render — Free Tier

Connect the repository as a Docker web service. Free instances sleep when idle and wake in roughly forty seconds — the threat-radar boot sequence covers the wake-up, exactly as designed.

NOTE

Restoring Live AI

The analysis uplink reads a provider config (baseUrl + apiKey) and speaks the OpenAI-compatible format — point it at any compatible provider, including local models, to upgrade verdicts from the Offline Analyst to live generation.

OPERATIONS NOTE

The first visit after a cold start triggers a full 19-source sync — a few seconds, covered by the boot sequence. Every visit afterwards is served from cache and feels instant. There is no cron, no worker and no maintenance: idle hosts simply start cold, and warm hosts stay warm on traffic.

Stay ahead of the breach.

CyberShield turns nineteen scattered research streams into one ranked, explainable, always-on operational picture — on hardware you control, with nothing to provision and nothing to pay for before it earns its keep.

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